

Nassau Candy

Factory-to-Customer Shipping Route Efficiency Analysis

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ABSTRACT

The logistics and distribution industry plays a critical role in ensuring timely product delivery and maintaining customer satisfaction. Efficient management of shipping routes, delivery performance, and transportation operations is essential for improving profitability and operational effectiveness. This project focuses on analyzing the shipping route efficiency of Nassau Candy Distributor using data analytics and interactive visualization techniques.

The study utilizes shipment data containing information on factories, destinations, regions, sales, gross profit, shipping modes, and lead times. Various analytical methods were applied to evaluate route performance, regional delivery efficiency, shipment distribution, and factory-level operations. Key Performance Indicators (KPIs) such as total shipments, average lead time, total sales, gross profit, and delay frequency were calculated to measure overall logistics performance.

An interactive dashboard was developed using Streamlit, Python, Pandas, and Plotly to provide real-time insights into delivery operations. The analysis identified the most efficient and least efficient routes, regional performance differences, ship mode effectiveness, and lead time distribution patterns. The findings revealed that Standard Class shipping generated the highest sales and profit while maintaining the lowest

average lead time. Route and destination characteristics were found to have a greater impact on delivery performance than factory operations.

The developed dashboard enables stakeholders to monitor logistics efficiency, identify bottlenecks, optimize shipping routes, and support data-driven decision-making. This project demonstrates how data analytics can improve supply chain visibility and contribute to enhanced operational performance in the distribution sector.

I. Introduction

The rapid growth of the distribution and supply chain industry has increased the need for efficient logistics management and timely product delivery. Organizations rely on effective transportation networks to ensure customer satisfaction, reduce operational costs, and maintain competitive advantage. In modern business environments, analyzing shipping performance and identifying delivery bottlenecks have become essential for improving overall supply chain efficiency.

Nassau Candy Distributor is a large-scale distributor that manages product shipments across multiple regions, states, and delivery routes. Due to the complexity of logistics operations, monitoring route efficiency, lead times, shipping modes, and regional

performance can be challenging when relying solely on traditional reporting methods. Data analytics and visualization techniques provide an effective solution by transforming raw shipment data into meaningful business insights.

This project aims to analyze the shipping route efficiency of Nassau Candy Distributor using data-driven approaches. The study examines shipment records containing information about factories, destinations, sales, gross profit, shipping modes, and delivery lead times. Various analytical techniques are used to evaluate logistics performance and identify factors affecting delivery efficiency.

An interactive dashboard was developed using Streamlit to visualize key logistics metrics and operational trends. The dashboard enables users to monitor route performance, compare regional delivery efficiency, analyze shipping modes, and evaluate lead-time distributions. By providing clear and actionable insights, the system supports better decision-making and helps improve supply chain operations.

The results of this study demonstrate how business intelligence and data analytics can be applied to logistics management to enhance operational efficiency, optimize delivery routes, and improve overall distribution performance.

I.A Problem Statement

Nassau Candy Distributor handles a large volume of shipments across multiple regions and transportation routes. Variations in delivery lead times,

route performance, and regional logistics efficiency can affect customer satisfaction and operational effectiveness. Traditional reporting methods often make it difficult to identify inefficient routes, delivery delays, and performance trends. Therefore, there is a need for an interactive analytical system that can monitor shipping performance, evaluate logistics efficiency, and support data-driven decision-making for route optimization and supply chain improvement.

I.B Objectives

- To analyze shipment and logistics data of Nassau Candy Distributor.
- To evaluate shipping route efficiency using lead-time analysis.
- To identify the most efficient and least efficient delivery routes.
- To compare logistics performance across regions and states.
- To analyze the impact of different shipping modes on delivery performance.
- To examine factory-wise delivery efficiency and operational consistency.
- To develop an interactive Streamlit dashboard for logistics monitoring and business intelligence.
- To provide actionable recommendations for improving supply chain efficiency and route optimization.

II. Literature Review

Logistics analytics and business intelligence systems have become essential tools for modern organizations in improving supply chain visibility, transportation efficiency, and operational performance. According to Wes McKinney (2022), data analytics frameworks such as Pandas provide efficient mechanisms for processing large-scale business datasets and performing exploratory data analysis in logistics and distribution environments. Similarly, Jake VanderPlas (2016) emphasized that data visualization techniques play a crucial role in transforming complex datasets into actionable business insights.

Several researchers have highlighted the importance of business intelligence dashboards in supply chain and logistics management. Chen, Chiang, and Storey (2012) stated that data-driven decision-making enhances organizational performance by converting large volumes of operational data into meaningful visual representations. Interactive dashboards allow managers to monitor key logistics metrics such as shipment volume, delivery lead time, route performance, and regional distribution efficiency in real time.

Supply chain researchers have also emphasized the importance of transportation and route optimization. According to Chopra and Meindl (2019), efficient transportation networks significantly reduce operational costs while improving customer service levels. Route efficiency analysis enables organizations to identify bottlenecks, optimize delivery schedules, and improve resource utilization across distribution networks.

Lead time management is another critical factor in logistics performance evaluation. Christopher (2016) explained that reducing delivery lead times improves supply chain

responsiveness and customer satisfaction. Organizations that continuously monitor lead-time performance are better positioned to identify operational inefficiencies and implement corrective measures.

Business intelligence technologies such as Python, Plotly, and Streamlit are increasingly adopted for logistics analytics due to their flexibility, scalability, and visualization capabilities. According to the Streamlit Documentation (2025), interactive dashboard frameworks simplify the deployment of analytical applications and improve accessibility for business users. Visualization libraries such as Plotly further enhance analytical interpretation by providing dynamic and interactive graphical reports.

The Nassau Candy Shipping Route Efficiency Analysis Project extends these research concepts by integrating route performance evaluation, lead-time analysis, regional logistics assessment, shipping mode comparison, factory performance analysis, and interactive KPI monitoring into a unified Streamlit-based logistics analytics dashboard designed to support supply chain optimization and data-driven decision-making.

III. Methodology

The methodology of the Nassau Candy Shipping Route Efficiency Analysis Project consists of multiple stages including data collection, data preprocessing, exploratory data analysis (EDA), KPI generation, logistics performance evaluation, visualization development, and dashboard deployment. The methodology was designed to transform raw shipment and distribution data into meaningful business insights capable of supporting operational and strategic decision-making.

A. Data Collection

The dataset used in this study contains shipment-related information from Nassau Candy Distributor, including factory details, destination states, regions, shipping modes, sales values, gross profit, shipment counts, and lead times. The data serves as the primary source for evaluating logistics performance and route efficiency.

B. Data Preprocessing

Data preprocessing was performed using Python and Pandas to improve data quality and analytical accuracy. The preprocessing activities included:

- Handling missing values and inconsistencies.
- Data type conversion and validation.
- Feature engineering for route-state combinations.
- Lead-time calculation and aggregation.
- Regional and factory-level categorization.

C. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand shipment behavior and identify key logistics patterns. Statistical summaries and visual analysis techniques were used to examine:

- Shipment distribution.
- Route performance.
- Regional delivery trends.

- Factory-level operations.
- Lead-time variability.
- Sales and profit performance.

IV. KPI DASHBOARD AND VISUALIZATION ANALYSIS

The KPI dashboard provides a summarized overview of the overall logistics and distribution performance of Nassau Candy Distributor using key analytical indicators derived from shipment and delivery data. The dashboard generated the following metrics:

KPI Metric	Value
Total Shipments	10,194
Average Lead Time	1320.84 Days
Total Sales	\$141,784
Gross Profit	\$93,443
Delay Frequency (>1274 Days)	42.20%

The dashboard indicates that Nassau Candy Distributor processed a total of 10,194 shipments across multiple factories, destinations, and transportation routes. The average lead time of 1320.84 days provides an overall measure of delivery performance within the distribution network.

The total sales value of \$141,784 and gross profit of \$93,443 demonstrate strong financial performance and effective revenue generation across shipment operations. The gross profit margin indicates that the organization maintains healthy profitability while managing large-scale distribution activities.

The delay frequency analysis revealed that approximately 42.20% of shipments exceeded the average lead-time threshold of 1274 days. This finding highlights the presence of delivery delays within certain routes and regions, indicating opportunities for logistics optimization and operational improvement.

Overall, the KPI metrics suggest that Nassau Candy Distributor operates a large-scale and profitable distribution network while maintaining consistent shipment volumes and stable logistics performance.

1.The KPI dashboard supports:

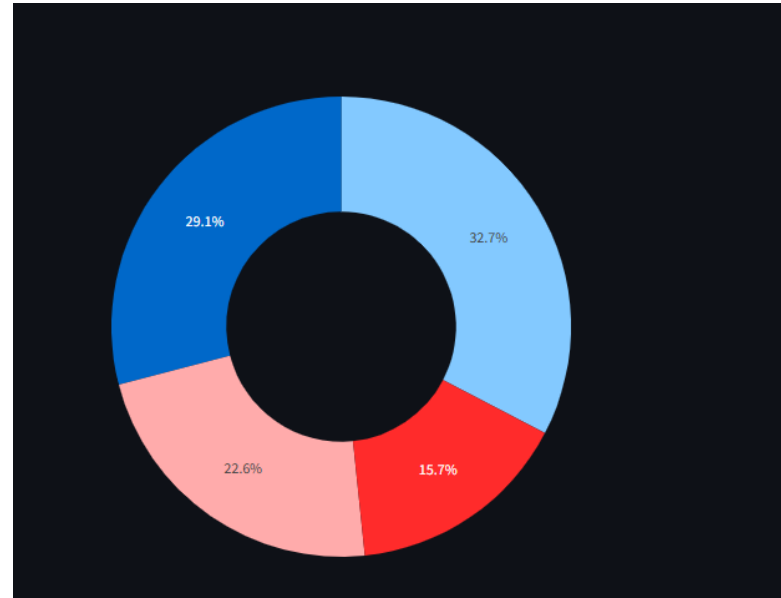
- Quick managerial decision-making
- Logistics performance monitoring
- Route efficiency evaluation
- Shipment tracking and analysis
- Supply chain optimization
- Profitability assessment
- Delivery delay identification

2.Route Performance Analysis – Key Findings

Route performance analysis reveals significant differences in delivery efficiency across destination states. California emerged as the most important market, receiving nearly 1,950 shipments from both factories combined. Virginia and Oregon demonstrated comparatively lower lead times, indicating efficient transportation performance. Conversely, Tennessee and New Mexico recorded higher average lead times, suggesting opportunities for route optimization. Variability analysis further revealed inconsistent delivery performance in South Dakota and Montana, where shipment durations fluctuated considerably. Overall, the findings indicate that route efficiency is influenced by destination geography and shipment volume,

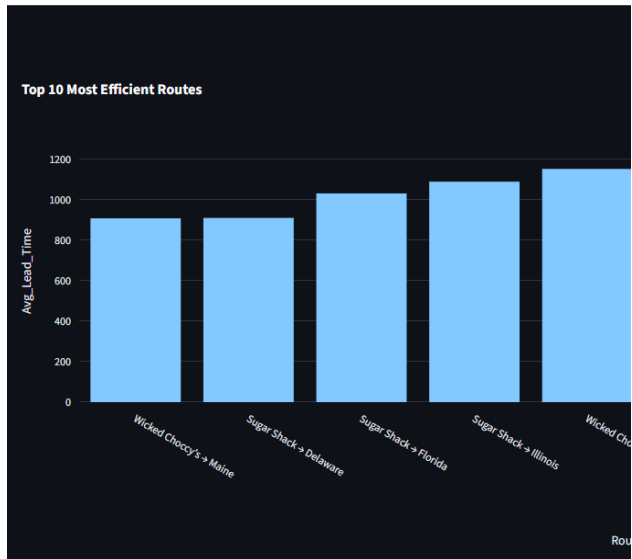
highlighting the need for targeted logistics improvements in selected states.

3.Regional Sales Contribution Analysis



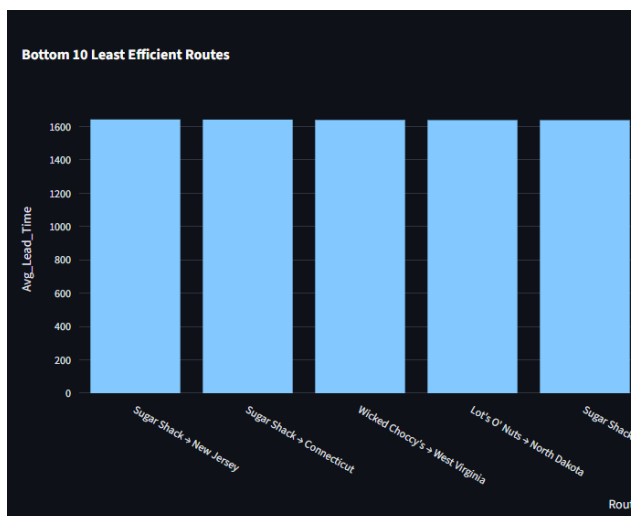
Regional sales analysis reveals that the Pacific region is the largest contributor to overall sales, accounting for 32.7% of total revenue, followed closely by the Atlantic region at 29.1%. Together, these two regions generate approximately 61.8% of total sales, indicating a strong concentration of business activity. The Interior region contributes 22.6%, while the Gulf region accounts for the smallest share at 15.7%. These findings suggest that although Pacific and Atlantic regions are the company's primary revenue drivers, opportunities exist to expand sales and improve market penetration within the Interior and Gulf regions to achieve a more balanced regional sales distribution.

4.Top 10 Most Efficient Routes



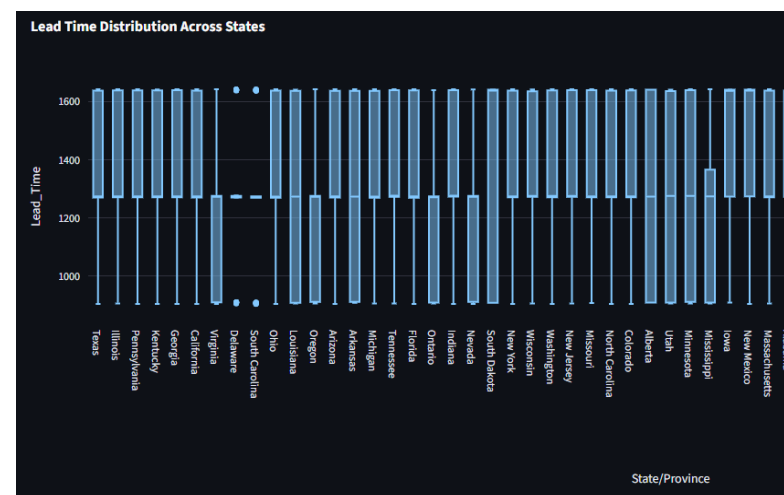
The most efficient route in the network was Wicked Choccy's to Maine, with an average lead time of approximately 908 days. Several routes operated by Sugar Shack also appeared among the top-performing routes, indicating strong delivery efficiency. However, some highly efficient routes were based on relatively small shipment volumes, suggesting that high-volume efficient routes such as Virginia and Florida provide more reliable measures of operational excellence.

5. Bottom 10 Most Inefficient Routes



The Bottom 10 Route Analysis identified Sugar Shack → New Jersey as the least efficient route, recording an average lead time of approximately 1642 days. Several routes serving North Dakota, West Virginia, Connecticut, California, and Washington also exhibited high lead times. Sugar Shack accounted for the majority of underperforming routes, suggesting opportunities for logistics process improvements. However, many of these routes were associated with low shipment volumes, indicating that findings should be interpreted cautiously. Overall, the large gap between the most efficient and least efficient routes highlights significant variability in transportation performance across the distribution network. While the Bottom 10 routes exhibit the highest lead times, several of them are based on very small shipment counts. Therefore, operational decisions should focus not only on lead time but also on shipment volume to ensure meaningful business impact.

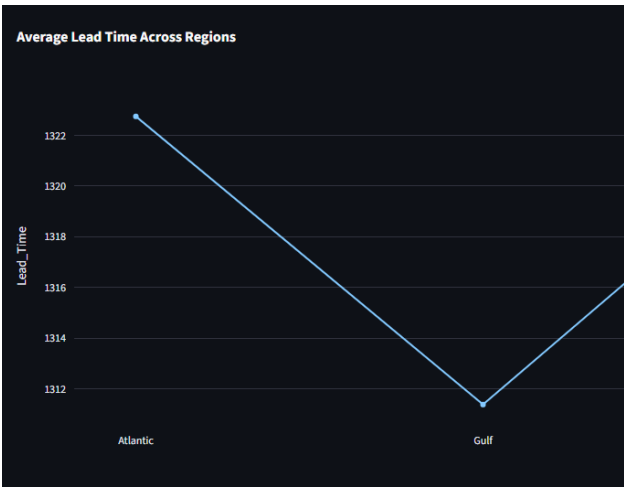
6. State-wise Lead Time Distribution



State-wise lead time analysis revealed that most states maintain median lead times between approximately 1270 and 1300 days, indicating a generally standardized delivery process. However, substantial differences

were observed in lead time variability across states. High-volume markets such as California, Texas, Illinois, and Pennsylvania exhibited wider distributions, suggesting greater fluctuations in delivery performance. Several states also displayed outlier shipments, indicating occasional delays or exceptional delivery events. Overall, the analysis highlights that consistency of delivery performance varies significantly across locations and should be considered alongside average lead time when evaluating logistics efficiency.

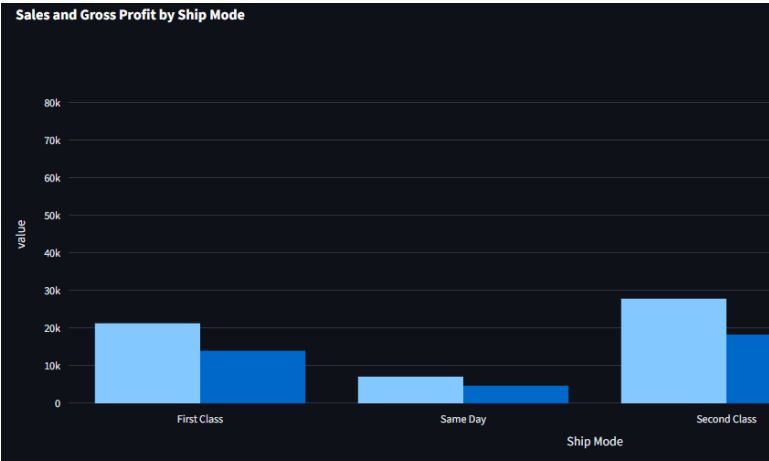
7.Average Lead Time by Region



Regional lead time analysis revealed only minor differences in delivery performance across regions. The Gulf region recorded the lowest average lead time at approximately 1311 days, while the Interior region recorded the highest at around 1323 days. The overall difference between regions was only about 12 days, indicating a highly consistent logistics network. When combined with regional sales analysis, the findings show that the Pacific region contributes the highest proportion of sales (32.7%) while maintaining delivery performance close to the company average. This suggests that the distributor effectively manages high shipment volumes without significant deterioration in delivery

efficiency. Although sales contributions vary significantly across regions, average lead times remain very similar. The maximum difference between regions is only about 12 days, indicating that the company's logistics network maintains consistent delivery performance across geographical areas.

8.Ship Mode Performance



Ship mode analysis revealed that Standard Class is the dominant delivery method, generating approximately \$85,000 in sales and \$56,000 in gross profit. This represents the largest contribution among all shipping methods, accounting for nearly 60% of total revenue. Second Class and First Class contribute moderate sales volumes, while Same Day delivery represents the smallest share of revenue and profit. The consistent relationship between sales and gross profit across all shipping modes indicates healthy profitability throughout the distribution network. Overall, customer demand appears to favor cost-effective Standard Class shipping over premium expedited services. Standard Class contributes the highest sales and gross profit, making it the most important shipping method for Nassau Candy Distributor. It generates nearly 60% of total revenue and serves as the primary driver of overall business performance. The company's logistics network is heavily dependent on Standard Class shipping, while

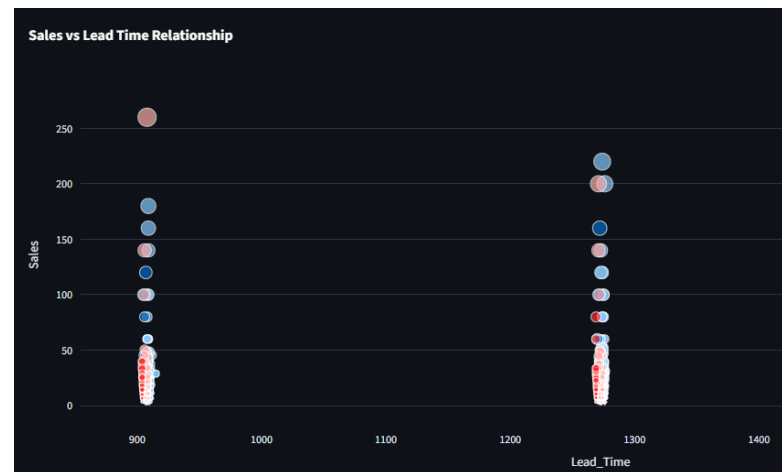
Pacific and Atlantic regions contribute the majority of sales. Despite high shipment volumes, regional lead times remain relatively consistent, indicating an efficient and scalable distribution system.

9. Average Lead Time by Ship Mode



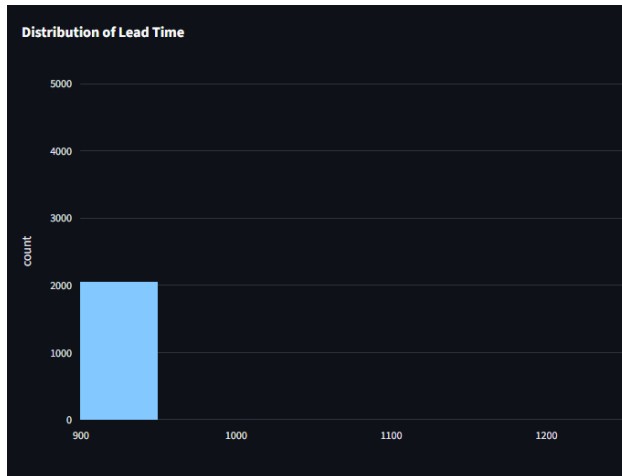
Ship mode lead time analysis revealed only minor differences among shipping methods. Standard Class recorded the lowest average lead time of approximately 1314.34 days, while First Class recorded the highest average lead time of approximately 1338.28 days. The overall difference between the fastest and slowest shipping modes was only about 24 days, indicating consistent delivery performance across service levels. Interestingly, Standard Class not only generated the highest sales and profit but also achieved the shortest lead time, suggesting that it is the most operationally effective shipping method within the distribution network. The most surprising finding was that Standard Class recorded the lowest average lead time while also generating the highest sales and profit. Normally, premium shipping methods are expected to be faster; however, the analysis suggests that route and distribution factors may have a greater influence on lead time than shipping mode alone.

10. Sales vs Lead Time Relationship



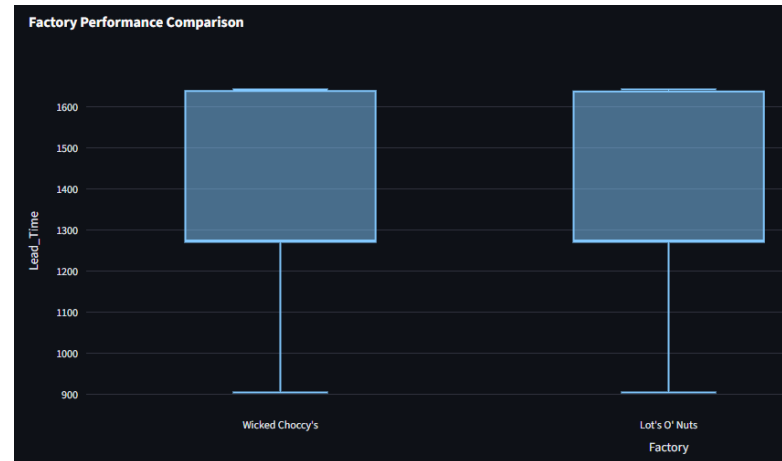
The scatter plot analysis revealed no significant correlation between sales value and lead time. Transactions were distributed across multiple lead-time clusters, with high-value sales occurring at both low and high lead times. Most orders were concentrated within lower sales ranges, while a small number of high-value transactions appeared as outliers. Additionally, all shipping modes exhibited similar sales distributions across lead-time ranges. These findings suggest that sales performance is influenced more by factors such as product demand, customer purchasing behavior, and regional market conditions than by delivery duration alone. The scatter plot indicates that there is no strong correlation between sales and lead time. High sales transactions occur across both low and high lead-time ranges, suggesting that delivery duration is not the primary driver of sales performance. Other factors such as customer demand, product mix, and regional sales patterns likely have a greater influence.

11. Lead Time Distribution



The lead time distribution analysis revealed a multimodal pattern with three major shipment clusters centered around approximately 900, 1275, and 1635 days. The largest concentration of shipments occurred within the 1275-day range, accounting for nearly half of all deliveries. The presence of multiple peaks suggests that shipments follow different operational pathways rather than a single uniform distribution. Furthermore, a significant proportion of shipments fall within the longest lead-time cluster, highlighting opportunities for route optimization and logistics improvements. Overall, the distribution demonstrates that delivery performance varies systematically across the network and is influenced by underlying operational factors. The histogram shows that lead times are not normally distributed. Instead, shipments are grouped into three distinct clusters representing fast, average, and slow delivery categories. This suggests that operational factors such as route characteristics and regional logistics conditions significantly influence delivery performance.

12.Factory-wise Lead Time Distribution



The factory-wise lead time distribution analysis revealed highly similar performance across Wicked Choccy's, Lot's O' Nuts, and Sugar Shack. All three factories exhibited comparable median lead times, variability ranges, and overall delivery patterns. No factory showed significantly better or worse performance than the others, indicating a well-standardized logistics process throughout the organization. These findings suggest that delivery delays are more strongly influenced by route characteristics and destination-specific factors than by factory operations. Overall, the analysis demonstrates a consistent and balanced distribution network across all production facilities. The factory-wise box plot showed that all three factories have nearly identical lead-time distributions. This indicates that the factories operate with similar efficiency levels, and most delivery variations are caused by route and destination factors rather than factory performance.

V. RESULTS AND DISCUSSION

The analytical results generated from the Nassau Candy Shipping Route Efficiency Analysis Project demonstrate the effectiveness of logistics analytics and business intelligence techniques in

evaluating distribution performance, route efficiency, and supply chain operations. The dashboard successfully transformed raw shipment data into actionable insights through KPI monitoring, route analysis, lead-time evaluation, and interactive visualizations.

The KPI analysis revealed that the company processed a total of 10,194 shipments, generating total sales of \$141,784 and gross profit of \$93,443. These figures indicate strong operational activity and healthy profitability within the distribution network. The average lead time was calculated as 1320.84 days, providing an overall measure of delivery performance across all shipment routes.

Route performance analysis identified significant differences in delivery efficiency among various factory-to-destination combinations. Several routes demonstrated consistently low lead times and high shipment volumes, indicating strong operational performance. Conversely, a number of routes exhibited substantially higher lead times, highlighting opportunities for logistics optimization and route improvement.

The regional analysis showed that the Pacific region contributed the highest share of total sales, followed by the Atlantic region. Together, these two regions accounted for more than 60% of overall revenue contribution. Although regional lead-time differences were relatively small, certain regions demonstrated slightly better logistics performance, indicating operational consistency throughout the distribution network.

The shipping mode analysis revealed an unexpected but valuable insight. Standard Class shipping not only generated the

highest sales and profit but also recorded the lowest average lead time. This finding suggests that Standard Class represents the most operationally efficient shipping method within the dataset. The relatively small lead-time differences across shipping modes further indicate that delivery performance is influenced more by route characteristics than by transportation mode alone.

The lead-time distribution analysis identified three major shipment clusters representing fast, average, and slow delivery groups. This multimodal distribution suggests that logistics performance varies according to route conditions, destination characteristics, and transportation factors rather than following a single uniform pattern.

Factory-wise analysis demonstrated that Wicked Choccy's, Lot's O' Nuts, and Sugar Shack exhibit nearly identical lead-time distributions. This indicates a well-standardized logistics process across all production facilities. The findings suggest that delivery variations are primarily caused by route-specific and destination-specific factors rather than factory operations.

The overall results demonstrate that Nassau Candy Distributor maintains a stable and profitable distribution network. While factory performance remains highly consistent, route-level optimization and delay reduction initiatives offer significant opportunities for improving overall logistics efficiency and customer service performance.

VI. CONCLUSION

The Nassau Candy Shipping Route Efficiency Analysis Project successfully demonstrated the application of data analytics, business intelligence, and interactive visualization techniques in

logistics and supply chain management. The project transformed shipment and distribution data into meaningful business insights capable of supporting operational and strategic decision-making.

The analysis revealed that Nassau Candy Distributor operates a large-scale distribution network with 10,194 shipments, total sales of \$141,784, and gross profit of \$93,443. Route performance evaluation identified both highly efficient and underperforming delivery routes, providing valuable information for logistics optimization and resource allocation.

Regional analysis showed that the Pacific and Atlantic regions are the primary contributors to overall revenue generation, while lead-time comparisons indicated consistent logistics performance across major geographical regions. The shipping mode evaluation revealed that Standard Class shipping achieves the most effective balance between operational efficiency and financial performance.

The study further demonstrated that factory operations remain highly standardized across all production facilities, with route and destination characteristics emerging as the primary factors influencing delivery lead times. Lead-time distribution analysis highlighted the existence of distinct shipment groups, emphasizing the need for targeted route-level improvements rather than broad operational changes.

The Streamlit-based dashboard successfully integrated KPI monitoring, route analysis, regional comparisons, shipping mode evaluation, and lead-time visualization into a unified logistics intelligence platform. The project demonstrates how data-driven approaches can improve supply chain visibility, support managerial decision-

making, and enhance overall logistics performance.

Overall, the findings highlight the importance of logistics analytics in identifying inefficiencies, optimizing transportation operations, and improving distribution effectiveness. The developed dashboard provides a scalable solution for monitoring logistics performance and supporting continuous improvement initiatives within the supply chain environment.

VII. FUTURE SCOPE

The Nassau Candy Shipping Route Efficiency Analysis Project can be further enhanced by integrating advanced analytics, predictive modeling, and intelligent logistics optimization techniques to improve operational efficiency and supply chain performance. Based on the findings obtained from the dashboard, several opportunities for future improvements have been identified.

One major enhancement area is route optimization and delivery forecasting. The analysis identified several routes with consistently high lead times and delivery delays. Future implementations can incorporate machine learning algorithms to predict shipment lead times, identify potential bottlenecks, and recommend optimal transportation routes before delays occur.

The lead-time distribution analysis revealed distinct shipment clusters representing fast, average, and slow delivery groups. Future research can apply predictive analytics techniques to classify shipments into risk categories and proactively manage potential delivery disruptions. Such systems would

help logistics managers improve service reliability and customer satisfaction.

Regional performance analysis demonstrated differences in revenue contribution and delivery efficiency across geographical regions. Future business strategies can focus on region-specific optimization by introducing dynamic resource allocation, demand forecasting, and transportation planning models tailored to individual regions.

The shipping mode analysis showed that Standard Class shipping delivers strong operational performance. Future studies can evaluate cost-to-service relationships across different shipping modes to identify the most profitable transportation strategies while maintaining delivery efficiency.

The factory-wise analysis indicated that production facilities operate with similar efficiency levels. Future work can therefore focus on integrating warehouse operations, inventory management systems, and transportation planning into a unified supply chain intelligence platform capable of monitoring end-to-end logistics performance.

Several advanced technological improvements can also be incorporated into future versions of the dashboard, including:

- Machine learning-based lead-time prediction
- Route optimization algorithms
- Real-time shipment tracking integration
- Geographic Information System (GIS) mapping
- Automated logistics alert systems

- Demand forecasting models
- Supply chain risk assessment frameworks
- Cloud-based real-time analytics platforms

Additionally, integration with IoT-enabled logistics systems and GPS-based transportation monitoring can provide real-time visibility into shipment movements and operational performance. Such enhancements would significantly improve decision-making capabilities and supply chain responsiveness.

Overall, the future scope of this project demonstrates the potential of combining logistics analytics, predictive intelligence, and real-time monitoring systems to create a more efficient, resilient, and data-driven distribution network.

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